

HARRY MAYNE

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I am an Astra Fellow working with Owain Evans (Truthful AI) and a PhD researcher at the University of Oxford with Adam Mahdi at [OxRML](#) and Jakob Foerster at [BOLD](#). My research focuses on LLM *self-explanations*: natural language explanations LLMs give to justify their own decision-making. I measure whether these explanations are faithful to models' true internal reasoning and I develop new training incentives to improve self-explanation faithfulness. I'm motivated by AI safety. My work has been published at leading venues including NeurIPS, ICML, and ICLR. Further research details can be found on my [website](#).

EDUCATION

University of Oxford

DPhil LLM Explainability and Interpretability

Oct. 2023 – Present

- Supervised by Prof. Adam Mahdi (Oxford Internet Institute) and Prof. Jakob Foerster (Engineering).
- Thesis on LLM explainability and interpretability, focusing on self-explanations.
- Fully funded by the Grand Union DTP (Economic and Social Research Council).
- Research agenda sponsored by the Dieter Schwarz Foundation.

University of Oxford

MSc Social Data Science

Distinction (77%)

Oct. 2022 – Aug. 2023

- Ranked 1st in cohort for overall exam performance and achieved the highest thesis mark (88%).
- Courses included Applied Machine Learning, Natural Language Processing, Data Analytics at Scale, Frontiers of Data Science and Applied Analytical Statistics.
- Fully funded by the Grand Union DTP (Economic and Social Research Council).

University of Cambridge

BA Economics

Double First Class

Oct. 2019 – Jun. 2022

- Ranked 1st/155 and 2nd/155 in econometrics and microeconomics examinations, respectively.
- Awarded the Patrick Cross Prize, two Tripos Prizes and the Corfield Scholarship for academic excellence.
- Courses included Time Series Econometrics, Microeconometrics, Labour Economics and Industrial Economics.
- **Thesis:** A Feast in the Time of Plague: A Theoretical Model of the UK Housing Market During COVID-19.

SELECTED PUBLICATIONS

2026	 Negation Neglect: When Models Fail to Learn Negations in Training. Mayne, H. [*] , McKinney, L. [*] , Dubinski, J., Karvonen, A., Chua, J., Evans, O.	Under Review at NeurIPS 2026
	 A Positive Case for Faithfulness: LLM Self-Explanations Help Predict Model Behavior. Mayne, H. [*] , Kang, J. [*] , Gould, D., Ramchandran, K., Mahdi, A., Siegel, N.	ICML 2026
	 LINGOLY-TOO: Disentangling Reasoning from Knowledge with Templatised Orthographic Obfuscation. Khouja, J., Korgul, K., Hellsten, S., Yang, L., Neacsu, V., Mayne, H., Kearns, R., Bean, A., Mahdi, A.	ICLR 2026

2025	🔗	LLMs Don't Know Their Own Decision Boundaries: The Unreliability of Self-Generated Counterfactual Explanations. Mayne, H., Kearns, R. O., Yang, Y., Delaney, E., Russell, C., Mahdi, A.	EMNLP 2025
	🔗	Ablation is Not Enough to Emulate DPO: A Mechanistic Analysis of Toxicity Reduction. Yang, Y., Sondej, F., Mayne, H., Lee, A., Mahdi, A.	EMNLP 2025
	🔗	Measuring What Matters: Construct Validity in Large Language Model Benchmarks. Bean, A., et al. (incl. Mayne, H.)	NeurIPS 2025
2024	🔗	LingOly: A Benchmark of Olympiad-Level Linguistic Reasoning Puzzles in Low-Resource and Extinct Languages. Bean, A., Hellsten, S., Mayne, H., Magomere, J., Chi, E., Chi, R., Hale, S., Kirk, H.	NeurIPS 2024, Oral, Top 0.5%
	🔗	Can Sparse Autoencoders be used to Decompose and Interpret Steering Vectors? Mayne, H., Yang, Y., Mahdi, A.	Interpretable AI, NeurIPS 2024

POSITIONS

Astra Fellowship: Owain Evans, Truthful AI	<i>Jan. 2026 – Jun. 2026</i>
Making sense of LLM generalization: Negation Neglect, (other work coming soon).	
SPAR Researcher: Noah Siegel, Google DeepMind	<i>Sept. 2025 – Apr. 2026</i>
- Developing a new explanatory faithfulness metric based on counterfactual simulatability. - First-author work at ICML 2026.	
International Growth Centre (IGC)	<i>Jun. 2025 – Present</i>
- AI Advisor to the Head of Data and AI (Jan. 2026 – Present). - AI Engineer (Jun. 2025 – Dec. 2025). - Using AI to aid public service delivery. See my IGC page.	
Bank of England	<i>Summer 2020 & 2021</i>
Internships in Insurance Supervision (PRA). Worked in <i>US Firms</i> and <i>Actuarial Science</i> teams.	

TEACHING

University of Oxford, Oxford Internet Institute	<i>Sep. 2023 – Dec. 2024</i>
- Teaching Assistant for Applied Analytical Statistics (Social Data Science MSc). - Taught over 50 Master's and PhD students. Voted the department's top TA of the year in 2024.	
Stanford University, Stanford House	<i>Nov. 2023 – Dec. 2025</i>
Tutored Stanford Computer Science undergraduates completing semesters abroad at Oxford. Mostly taught machine learning courses similar to CS229.	

ADDITIONAL SKILLS AND INTERESTS

Programming Languages: Python (PyTorch, vLLM, verl, Sklearn, Matplotlib, etc), L^AT_EX.

Conference Reviewing: NeurIPS 2024, 2025, 2026. ICLR 2026. ICML 2026.

Soft Skills: Public speaking, organisation, leadership, presentation.

Leadership Roles: Reasoning with Machines Lab Chair (Sept. 2025 – Dec. 2025).

Other Interests: Travel, surfing, and music.